A Bayesian Phylogenetic Analysis of the Nakh–Daghestanian Family on the Nichols 2003 Cognate Sets

Ekaterina Akhmetshina

July 2026

Abstract

I submit the cognate data from the Appendix 2 of Nichols 2003 to Bayesian Markov chain Monte Carlo (MCMC) phylogenetic inference, implemented in BEAST 2, through the BEASTling toolkit in order to test whether the internal structure of the Nakh–Daghestanian family can be recovered computationally from this small cognate dataset. The input is inherently sparse: 50 reconstructed proto-forms across 27 languages, with an average of roughly twelve languages per concept. I compare different configurations, but none of the them recovers the family's basic sub-groupings with meaningful support. I attribute this result to the scarcity of the data.

1 Introduction

The internal structure of the Nakh–Daghestanian language family remains only partially resolved. There is broad consensus on the family's two primary branches – a Nakh branch (Chechen, Ingush, Bats) and a larger Daghestanian branch – and on the reality of the major Daghestanian subgroups: Avar–Andi–Tsez, Lak, Dargi, Lezghian and Xinalug. The classical account I take as my reference is that of Nichols [5], who reconstructs Proto-Nakh–Daghestanian through regular sound correspondences; the consensus classification down to the level of these major branches is reproduced in Figure 1. Below that level, however, several internal relationships remain disputed. Ideally, computational historical linguistics methods could help resolve this uncertainty.

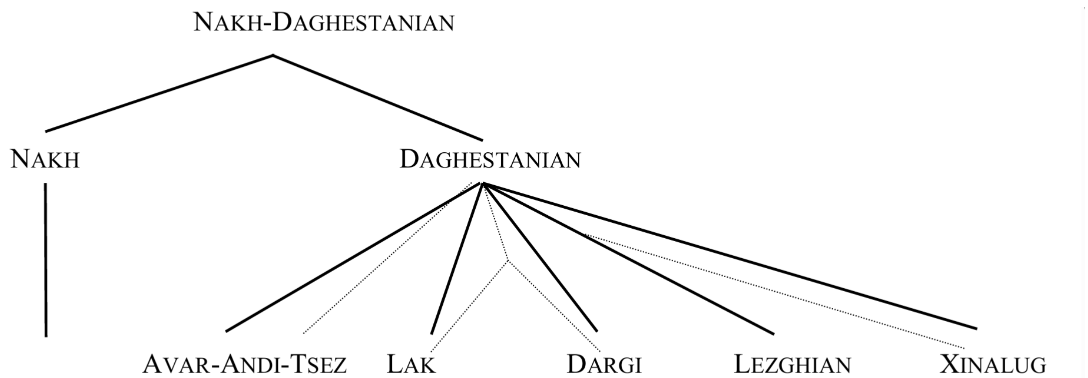

Figure 1: *Nakh-Daghestanian family tree. Major branches only. Light grey lines mark alternative subgroupings.*

Figure 1: The consensus classification of Nakh–Daghestanian, major branches only, after Nichols (2003). Light grey lines mark alternative subgroupings. This is the reference topology against which results are compared.

In this paper I am trying to answer the question whether it is possible to recover the structure of the family tree on the data, collected from cognate sets assembled in the appendix of Nichols 2003, and if it is possible, how precise can we get. I set out three predictions, ordered by ambition.

1. **The primary split.** The internal coherence of the Nakh languages (Chechen, Ingush, Bats) should be strong enough that even sparse data recover them as a clean clade, distinct from Daghestanian.

2. **The Daghestanian sub-branches.** The algorithm should further separate the major internal branches of Daghestanian. It would be interesting to see whether phylogenetic analysis on Nichols data reproduces the grouping that she established manually.

3. **Further resolution.** A result exceeding all my expectations would be a subdivision below the level of the recognised branches. Were I to reach this stage, the appropriate next step would be to test whether the additional structure is statistically robust and genuinely informative for the field.

However, I want to underline again that the dataset is small and was not assembled for phylogenetic purposes, so strong conclusions about the family tree are not to be expected. As will become clear at the end of this work, concerns about the size of the dataset are not unfounded.

2 Data

The sole source of data is Appendix 2 ("Selected PND Cognate Sets") of Nichols [5], which lists 50 cognate sets, one per reconstructed Proto-Nakh–Daghestanian form, together with the attested daughter forms in different languages of the family. Not every language is attested in every set.

I have digitized the data and saved it in CLDF format, which is used for phylogenetic analysis. Where a single set listed more than one form for the same language, I retained only the first one for simplification purposes and because CLDF format does not allow commas.

Even before analysis it was clear that 50 concepts over 27 languages is a dataset that is not merely sparse but also uneven, since the data were originally gathered to support a reconstruction rather than a phylogeny. Two tables in the appendix show the coverage in both directions: concepts attested per language (Table 1) and languages attested per concept (Table 2). Coverage is highly unequal. Chechen and Ingush are attested for all 50 concepts, whereas Tab, Ginux, STab and Budux fall below ten. On the concept side, the best-attested item ("apple") reaches only 20 of 27 languages, while the sparsest ("break", "what") reach just six; the average concept is attested in roughly twelve languages. I flagged languages attested for fewer than 15 concepts, and the most poorly attested concept sets under the assumption that these entries are most likely to contribute noise rather than signal.

However, two considerations pull against each other. On the one hand, the dataset is already small, so discarding information comes at a high cost. On the other hand, I assume that concept sets with very few attested languages carry little phylogenetic weight and may simply introduce noise. This is why I run the analysis both on the full set and on a reduced version obtained with BEASTLING's `minimum_data` filter, which drops any language attested for fewer than a set percentage of characters (Section 4).

3 Methods

Inference was carried out in BEAST 2, configured through BEASTLING, which runs Bayesian MCMC over tree topologies, branch lengths and model parameters given a cognate matrix. The choices below follow from the structure of the data rather than from convention.

Substitution model. The convention would be to use a covarion model, since covarion processes often fit lexical data well by allowing characters to switch between fast- and slow-evolving hidden states. I am also comparing it with Bayesian Stochastic Variable Selection (BSVS) binary model, which offers less flexibility, and therefore could be more suitable for already sparse data.

Clock model. A strict clock assumes a single rate of lexical replacement across all branches. This is almost certainly wrong for Nakh–Daghestanian, whose branches are expected to differ in rate. A fully relaxed clock, however, introduces up to $2n-2=52$ branch-rate parameters for 27 languages, in addition to the mean clock rate and the standard deviation of the log-normal distribution. With only 50 characters and heavy missingness this is dangerously close to over-parameterisation, and a relaxed clock would need reliable calibrations to converge. I therefore treat the strict clock as the baseline and reserve the relaxed clock for the stage of the protocol where a stable topology has already emerged.

Monophyly constraints. As a separate lever, I can impose the known top-level grouping as a prior (`monophyly`). Constraining the family at level 1 seeds the fundamental Nakh vs. Daghestanian bifurcation and lets us ask whether fixing the deepest split allows the data to resolve structure beneath it.

Chain length. Because the signal is weak, I run the MCMC longer than a dataset of this nominal size would ordinarily require. All runs reported here use a chain length of 20 M generations, with the first 10% discarded as burn-in.

Protocol. These choices are combined into the staged design shown in Figure 2. I first run BSVS and covarion under a strict clock and select the better-behaved model. The next step is contingent on the outcome: if the runs recover a reasonable Nakh–Daghestanian split, I escalate to a relaxed clock with rate variation and then repeat on the reduced matrix (`minimum_data = 20%`); if no reasonable split emerges, I instead impose monophyly at level 1 and test whether seeding the top split resolves the internal branches. Posterior consensus trees are visualised in SplitsTree.

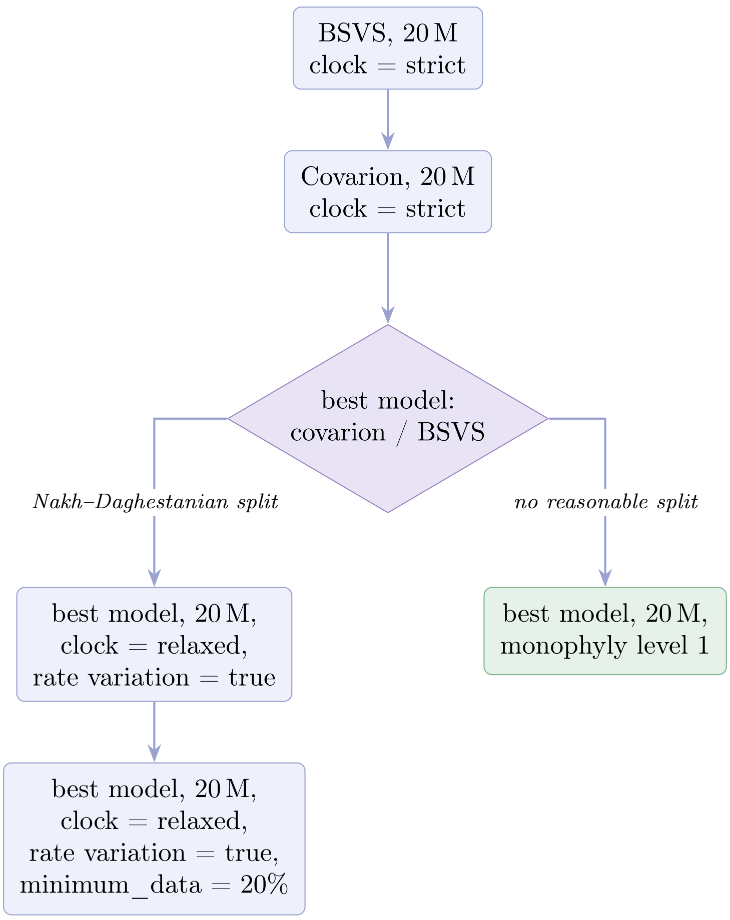

Figure 2: The staged experimental design. Model choice is settled first under a strict clock; the branch taken thereafter depends on whether the fundamental Nakh–Daghestanian split is recovered.

4 Results

4.1 Model comparison under a strict clock

The two strict-clock runs are shown in Figures 3 and 4. The great majority of languages are left unresolved: both trees are effectively flat rakes, with almost no supported internal structure.

Under the BSVS model the only resolved node with non-trivial support is a Dargi + Akhvakh grouping (posterior probability ≈ 66.7), which also appears under the covarion model as a nested clade (posterior probability ≈ 65.6). Under the covarion model Xvarshi is placed as the basal-most taxon, with a posterior probability of 100 for that root split.

Neither the Dargi + Akhvakh grouping nor a basal Xvarshi coincides with recognised groupings. One might read these as linguistically motivated, but in my view both are artefacts of the sparse signal. Most importantly, neither run recovers the fundamental Nakh–Daghestanian split. Following the protocol of Figure 2, the absence of a reasonable split directs us to the monophyly branch rather than to the relaxed clock, since the near-total lack of resolution makes it pointless to escalate model complexity. There were no meaningful differences in the performance of BSVS and covarion models, so I proceeded using BSVS.

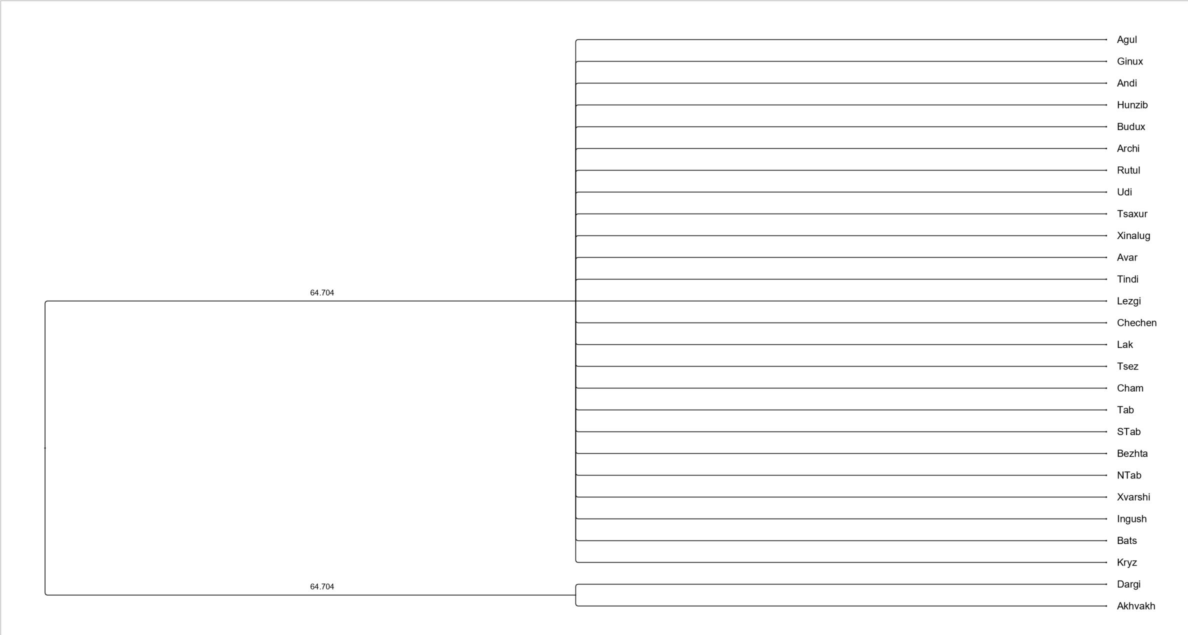

Figure 3: BSVS model, strict clock, 20 M generations. Only the two basal splits carry labels (≈ 66.7); internal structure is unresolved.

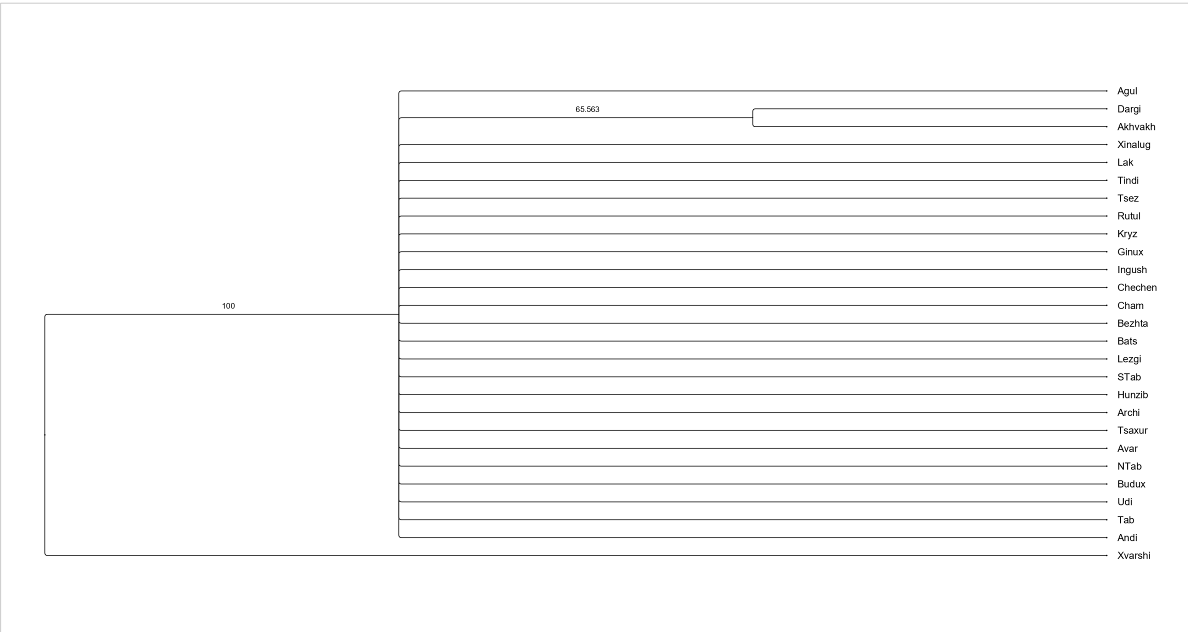

Figure 4: Covarion model, strict clock, 20 M generations. The tree is essentially flat; the only labelled internal grouping is Dargi + Akhvakh (≈ 65.6), with Xvarshi placed as basal-most taxon (root split $= 100$).

4.2 Seeding the top split: monophyly at level 1

Since the data alone did not recover the primary bifurcation, I decided to test whether supplying it as a constraint would let the internal branches resolve. Constraining monophyly at level 1 (Figure 5) recovers the Nakh vs. Daghestanian split by construction, at posterior probability 100, but the resolution does not propagate downward: within each constrained subfamily the topology remains an unresolved rake. In other words, fixing the deepest node did not help recover the further sub-branches. Considering that the data was insufficient to perform even the most basic split, this outcome is not all that surprising.

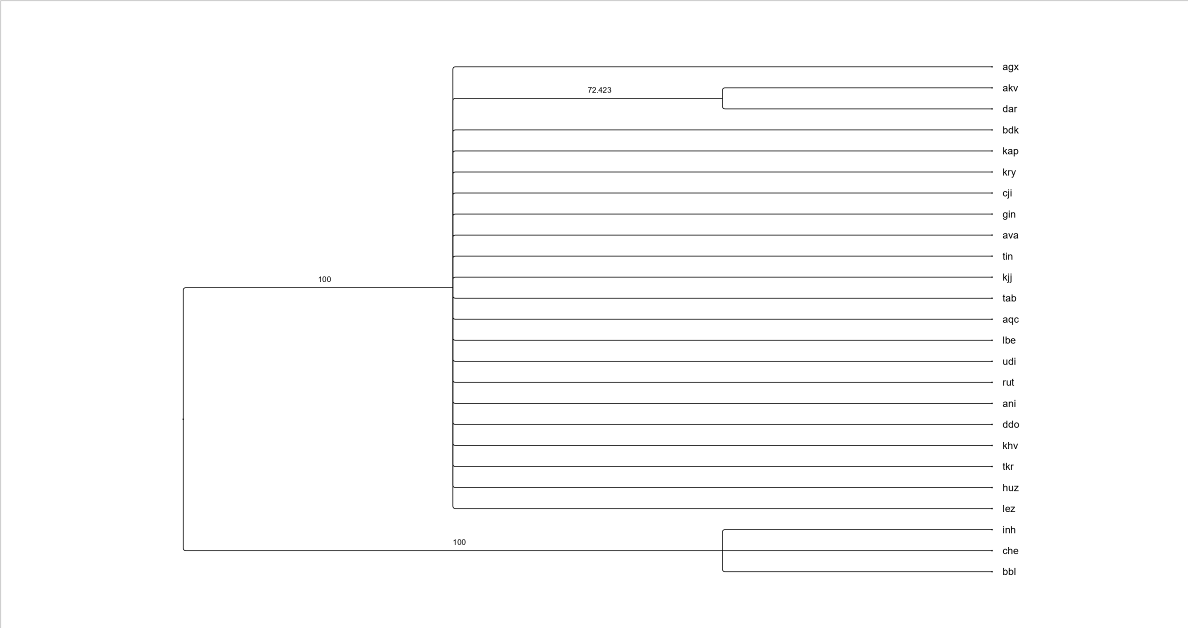

Figure 5: BSVS, strict clock, 20 M, with monophyly imposed at level 1. The constrained root split is recovered (posterior probability 100), but internal structure within each subfamily stays flat.

4.3 Cutting out the sparsest languages

I ran this test not as part of the protocol, but as a robustness check. Using `minimum_data = 20%` to exclude any language attested for fewer than 20% of the 50 characters – that is, fewer than ten concepts – which removes the most poorly documented languages (Tab, Ginux, STab, Budux), I repeated the BSVS, strict clock run.

The outcome (Figure 6) is unchanged in character: removing the sparsest languages does not help sharpen the internal topology, which remains flat apart from the previously attested Dargi + Akhvakh grouping (≈ 66.5).

Therefore the problem is the total quantity of signal, not a few especially thin rows.

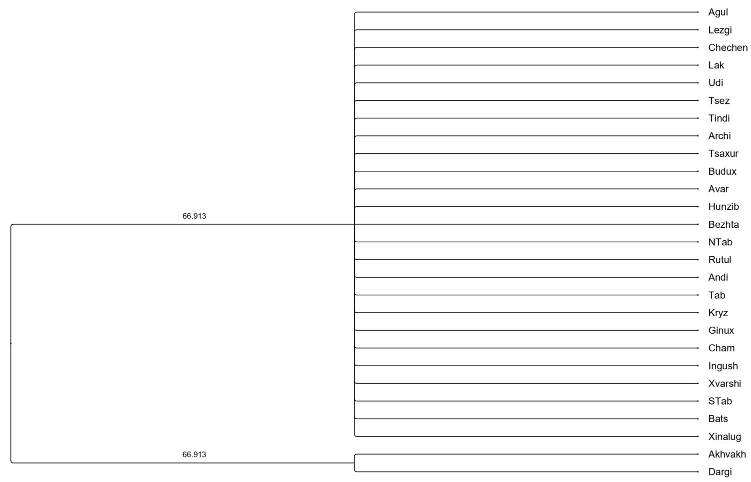

Figure 6: BSVS, strict clock, 20 M, reduced matrix (`minimum_data = 20%`). Trimming the sparsest languages leaves the topology essentially unresolved.

5 Conclusion and Limitations

The overall result is negative. Across every configuration – two substitution models, a monophyly constraint, and a reduced set of languages – the analysis fails to recover the major subgroupings of Nakh–Daghestanian with meaningful support. Without monophyly constraints it does not even recover the fundamental Nakh vs. Daghestanian split. The few labelled nodes that do appear are not linguistically interpretable.

Interpreting the flat topology. There are different reasons that could lead to a rake-like posterior. It could mean (i) that the MCMC has not converged, so the result is an artefact of incomplete sampling; (ii) that the chains have converged but the data carry too little signal to resolve the trees properly; or (iii) that the actual family tree truly looks like that, having split into many independent branches simultaneously.

In contemporary literature on the Nakh-Dagestani languages, doubts have indeed been raised regarding the unambiguous nature of the primary division into Nakh and Dagestani languages [2]. A possible alternative proposed is that all branches split from the root simultaneously. Nevertheless, the internal cohesion of language groups such as the Lezgian languages is indisputable [3], and our analysis does not reveal this. The flat topology reflanguages our inability to recover real structure from this data, not the absence of that structure, so reading (iii) can be rejected.

Data, not computation. Reading (i) can also be largely excluded, and here the argument is a priori rather than empirical. The strict-clock runs on which my conclusions rest have a small parameter space (a single clock rate and no per-branch rate parameters) so they are expected to mix well even at a chain length of 20 M. What does the work is the lack of data. With 50 binary characters and an average of about twelve attested languages per character, the total observable change is tiny: even if every cognate set changed state exactly once, that is at most 50 events over some 52 branches – under one substitution per branch – and the number of parsimony-informative characters, those able to discriminate between competing splits, is even smaller once missingness is taken into account. A matrix this thin cannot resolve a tree of this depth, and, consistently, forcing the deepest split does not help, because this constraint fixes a node without adding characters. Trimming the sparsest languages also does not help, because the deficit is in the aggregate signal rather than in a few thin rows.

The predictions, revisited. None of the three predictions in Section 1 have been confirmed. Prediction 1 (the Nakh languages form a pure cluster) is not reproduced in any run without restrictions: there is so little data that even the internal coherence of the Nakh languages is not strong enough to overcome this limitation.

Prediction 2 (separation of the Daghestanian subgroups) is also not confirmed; the analysis does not reproduce any of the branches shown in Figure 1 unless they are specified in advance, and when the top-level division is specified using a monophyly constraint, the resolution does not propagate downward. Prediction 3 (novel finer-grained structure) is even less supported, and the question of statistical significance, which it implied, does not even arise.

Thus, the following conclusions can be drawn: to obtain any meaningful computational results for this family, much more data needs to be collected. Such work goes beyond the scope of this study, and it seems to me that available data alone may not be sufficient and that fieldwork will be necessary.

After gathering the data, the next step would be to figure out whether any computational results could address the open questions in the family classification.

Data and Code Availability

The cognate dataset, the BEASTLING configuration files and the coverage tables are available in the project repository: nd-cognates.

References

[1] Bouckaert, R., Heled, J., Kühnert, D., Vaughan, T., Wu, C.-H., Xie, D., Suchard, M. A., Rambaut, A., & Drummond, A. J. (2014). BEAST 2: A software platform for Bayesian evolutionary analysis. PLOS Computational Biology 10(4): e1003537.

[2] Ganenkov, D., & Maisak, T. (2021). Nakh-Dagestanian languages. In M. Polinsky (ed.), The Oxford Handbook of Languages of the Caucasus. Oxford: Oxford Academic.

[3] Kassian, A. S. (2017). Linguistic homoplasy and phylogeny reconstruction: The cases of Lezgian and Tsezic languages (North Caucasus). Folia Linguistica 51(s38–s1): 217–262.

[4] Maurits, L., Forkel, R., Kaiping, G. A., & Atkinson, Q. D. (2017). BEASTling: A software tool for linguistic phylogenetics using BEAST 2. PLOS ONE 12(8): e0180908.

[5] Nichols, J. (2003). The Nakh-Daghestanian consonant correspondences. In D. A. Holisky & K. Tuite (eds.), Current Trends in Caucasian, East European and Inner Asian Linguistics: Papers in Honor of Howard I. Aronson, 207–264. Amsterdam: John Benjamins.

A Coverage Tables

Table 1: Number of cognate sets (of 50) attested for each language. Languages attested for fewer than 15 concepts are marked with †.

Language	Concepts	Language	Concepts
Agul	38	Lak	41
Akhvakh	22	Lezgi	26
Andi	34	NTab	34
Archi	38	Rutul	31
Avar	40	STab†	6
Bats	34	Tab†	1
Bezhta	17	Tindi	23
Budux†	7	Tsaxur	16
Cham	27	Tsez	24
Chechen	50	Udi	31
Dargi	39	Xinalug	32
Ginux†	3	Xvarshi†	10
Hunzib	18		
Ingush	50		
Kryz	27		

Table 2: Number of languages (of 27) attested for each concept set. The sparsest sets, attested in six languages, are marked with †.

Concept	Langs	Concept	Langs	Concept	Langs
apple	20	fire	16	pig	12
ashes	15	five	12	ram	8
axe	7	frost	10	right	11
back	15	hand	11	road	13
barley	13	heart	17	round	16
bear	17	know	8	sun	18
bile	9	leave	14	take	9
black	9	long	17	thin	14
break†	6	measure	10	tooth	15
bull	10	meat	17	tongue	18
burn	9	moon	18	water	18
die	13	name	18	we	13
dry	17	nettle	17	weave	10
earth	11	one	17	what†	6
eye	13	pear	12	winter	9
fill	14			wool	8
				year	16
				yoke	15